3.6.2 Roots of Quadratic Equations

Let $ax^2 + bx + c = 0, (a \neq 0)$ be a quadratic equation. The values of x such that the expression $ax^2 + bx + c$ becomes zero are called roots of the quadratic equation $ax^2 + bx + c = 0$.

We have, $ax^2 + bx + c = 0$

$$a\left[x^2 + \frac{b}{a}x + \frac{c}{a}\right] = 0$$

$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0 \quad \left(\text{since } a \neq 0\right)$$

$$x^2 + \frac{b}{2a}(2x) + \left(\frac{b}{2a}\right)^2 - \left(\frac{b}{2a}\right)^2 + \frac{c}{a} = 0$$

That is, $$x^2 + (2x)\left(\frac{b}{2a}\right) + \left(\frac{b}{2a}\right)^2 = \frac{b^2}{4a^2} - \frac{c}{a}$$

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}$$

$$x + \frac{b}{2a} = \frac{\pm\sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Therefore, the roots are $\frac{-b + \sqrt{b^2 - 4ac}}{2a}$ and $\frac{-b - \sqrt{b^2 - 4ac}}{2a}$

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3.6.3 Formation of a Quadratic Equation

If α and β are roots of a quadratic equation $ax^2 + bx + c = 0$ then

$$\alpha = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \text{ and } \beta = \frac{-b - \sqrt{b^2 - 4ac}}{2a}.$$

Also, $$\alpha + \beta = \frac{-b + \sqrt{b^2 - 4ac} - b - \sqrt{b^2 - 4ac}}{2a} = \frac{-b}{a}$$

and $$\alpha\beta = \left(\frac{-b + \sqrt{b^2 - 4ac}}{2a}\right) \times \left(\frac{-b - \sqrt{b^2 - 4ac}}{2a}\right) = \frac{c}{a}.$$

Since, $(x - \alpha)$ and $(x - \beta)$ are factors of $ax^2 + bx + c = 0$,

We have $$(x - \alpha)(x - \beta) = 0$$

Hence, $$x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

That is, $x^2 -$ (sum of roots) $x +$ product of roots $= 0$ is the general form of the quadratic equation when the roots are given.

Note

$ax^2 + bx + c = 0$ can equivalently be expressed as $x^2 + \frac{b}{a}x + \frac{c}{a} = 0$. since a≠0

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